

LEVEL - III

- 100 boys are randomly divided into two groups containing 50 boys each. The probability that the two tallest boys are in different groups is
1) $50/99$ 2) $49/99$ 3) $25/99$ 4) $1/2$
- Let $x = 33^n$. The index n is given a positive integral value at random. The probability that the value of x will have 3 in the units place is
1) $1/4$ 2) $1/2$ 3) $1/3$ 4) $1/5$
- Consider all functions that can be defined from the set $A = \{1, 2, 3\}$ to the set $B = \{1, 2, 3, 4, 5\}$. A function $f(x)$ is selected at random from these functions. The probability that selected function satisfies $f(i) \leq f(j)$ for $i < j$ is equal to
1) $6/25$ 2) $7/25$ 3) $2/5$ 4) $12/25$
- Three children are selected at random from a group of 6 boys and 4 girls. It is known that in this group exactly one girl and one boy belong to same parents. The probability that the selected group of children have no blood relations, is equal to
1) $11/15$ 2) $13/15$ 3) $14/15$ 4) $2/15$
- A biased ordinary die is loaded in such a way that probability of getting an even outcome is five times the probability of getting an odd outcome. This die is rolled two times. The probability that the sum of outcomes will be a prime number, is equal to
1) $67/324$ 2) $63/324$ 3) $123/324$ 4) $71/324$
- In a convex hexagon two diagonals are drawn at random. The probability that the diagonals intersect at an interior point of the hexagon is
1) $\frac{5}{12}$ 2) $\frac{7}{12}$ 3) $\frac{1}{26}$ 4) $\frac{2}{51}$
- A sum of money is rounded off to the nearest rupee. The probability that error occurred in rounding off is at least 15 paise is
1) $\frac{29}{101}$ 2) $\frac{29}{100}$ 3) $\frac{71}{101}$ 4) $\frac{71}{100}$
- If the integers 'm' and 'n' are chosen at random from 1 to 100 then the probability that a number of the form $7^m + 7^n$ is divisible by 5 is
1) $\frac{1}{5}$ 2) $\frac{1}{7}$ 3) $\frac{1}{4}$ 4) $\frac{1}{49}$

- If $a \in [-6, 12]$, the probability that graph of $y = -x^2 + 2(a + 4)x - (3a + 40)$ is strictly below x-axis is
1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$
- Seven chits are numbered from 1 to 7. Four are drawn one by one with replacement. The probability that the least number on any selected chit is 5 is
1) $1 - \left(\frac{2}{7}\right)^4$ 2) $4 - \left(\frac{2}{7}\right)^4$ 3) $\left(\frac{3}{7}\right)^4 - \left(\frac{2}{7}\right)^4$ 4) $\left(\frac{3}{7}\right)^4$
- If four whole numbers taken at random are multiplied together, then the probability that the last digit in the product is 1, 3, 7 or 9 is
1) $\frac{16}{625}$ 2) $\frac{32}{625}$ 3) $\frac{64}{625}$ 4) $\frac{256}{675}$
- A natural number x is chosen at random from the first 1000 natural numbers. If $[.]$ denotes the greatest integer function then the probability that $\left[\frac{x}{2}\right] + \left[\frac{x}{3}\right] + \left[\frac{x}{5}\right] = \frac{31x}{30}$ is
1) $33/1000$ 2) $34/1000$ 3) $32/1000$ 4) $31/1000$
- Given two events A and B, if the odds against A are 2 to 1, and those in favour of $A \cup B$ are 3 to 1, then
1) $\frac{1}{3} \leq P(B) \leq \frac{1}{2}$ 2) $\frac{1}{2} \leq P(B) \leq \frac{3}{4}$
3) $\frac{5}{12} \leq P(B) \leq \frac{3}{4}$ 4) $0 \leq P(B) \leq 1$
- There are ninety cards in a box numbered 10, 11, 12, ..., 98, 99. Three cards are drawn from the box one by one with replacement. The probability that product of the digits on the cards will be 12 at least once is
1) $1 - \left(\frac{4}{5}\right)^3$ 2) $\left(\frac{43}{45}\right)^3$ 3) $1 - \left(\frac{43}{45}\right)^3$ 4) $\frac{43}{45}$
- Every evening a student either watches TV or reads a book. The probability of watching TV is $\frac{4}{5}$. If he watches TV, the chance that he will fall a sleep is $\frac{3}{4}$ and it is $\frac{1}{4}$ when he read a book. On one evening, the student is found to be asleep. The probability that he watched TV is
1) $\frac{2}{13}$ 2) $\frac{12}{13}$ 3) $\frac{11}{13}$ 4) $\frac{1}{13}$

16. α is a solution of the equation $z^n = (z+1)^n$ where $n \geq 2, n \in \mathbb{N}$. Then the probability that α lies on the real axis is
 1) $\frac{1}{n}$ 2) $\frac{2}{n}$ 3) $\frac{1}{n-1}$ 4) $\frac{2}{n-1}$
17. A critical point x_0 of the function $f(x) = x^3$ is selected at random. The probability that f is extremum at x_0 is
 1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) 0 4) 1
18. An ellipse of eccentricity $\frac{2\sqrt{2}}{3}$ is inscribed in a circle and a point within a circle is chosen at random. Then the probability that this point lies outside the ellipse is
 1) $\frac{2}{3}$ 2) $\frac{8}{9}$ 3) $\frac{1}{3}$ 4) $\frac{2}{5}$
19. Two persons A and B agree to meet at a place between 11 to 12 noon. The first one to arrive waits for 20 minutes and then leave. If the time of their arrival be independent and at random, The probability that A and B meet
 1) $\frac{4}{9}$ 2) $\frac{5}{9}$ 3) $\frac{2}{9}$ 4) $\frac{1}{9}$
20. Two numbers x and y are chosen at random (without replacement) from amongst the numbers $1, 2, 3, \dots, 3n$. Then the probability that $x^3 + y^3$ is divisible by 3 is
 1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$
21. Two non-negative integers are chosen at random from the set of non-negative integers with replacement. The probability that the sum of their squares is divisible by 10 is
 1) $\frac{3}{50}$ 2) $\frac{6}{50}$ 3) $\frac{9}{50}$ 4) $\frac{12}{50}$
22. Let F be the set of all 4 digit numbers whose sum is 34. If a number is selected from F , the probability that the selected number is even is
 1) $\frac{1}{10}$ 2) $\frac{2}{10}$ 3) $\frac{3}{10}$ 4) $\frac{4}{10}$
23. A fair coin is tossed 100 times. The probability of getting tails 1, 3, ..., 49 times is
 1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{8}$ 4) $\frac{1}{16}$
24. If two numbers selected from $1, 2, 3, \dots, 25$ and 'P' is the probability that difference between them is less than 10 then the value of '5P' is
 1) 1 2) 2 3) 3 4) 4
25. If a and b are chosen randomly from the set consisting of numbers $1, 2, 3, 4, 5, 6$ with replacement then the probability that $\lim_{x \rightarrow 0} \left[(a^x + b^x) / 2 \right]^{2/x} = 6$ is
 1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{1}{9}$ 4) $\frac{2}{9}$
26. An experiment has 10 equally likely outcomes. Let A and B be two non empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is
 1) 2, 4 or 8 2) 3, 6 or 9 3) 4 or 8 4) 5 or 10
27. In a test of MCQ with 4 choices a student either guess or remember or computes the answer. Probability of making a guess is $\frac{1}{3}$, remembering is $\frac{1}{6}$ and giving correct answer (when remembered) is $\frac{1}{8}$, probability of computing correct answer is
 1) $\frac{11}{24}$ 2) $\frac{23}{24}$ 3) $\frac{23}{29}$ 4) $\frac{24}{29}$
28. The probabilities of Ramesh using car or scooter or bus or train for going to office are respectively $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$. The probabilities of his reaching the office late using these modes of transport are respectively $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$. On one day Ramesh reaches his office on time. Then the probability that he used car on that day is
 1) $\frac{1}{8}$ 2) $\frac{1}{7}$ 3) $\frac{1}{6}$ 4) $\frac{2}{5}$

29. Three dice are thrown, the numbers appearing on them are respectively a, b and c . chance that the roots of the quadratic equation $ax^2 + bx + c = 0$ are real is
- 1) $\frac{43}{216}$ 2) $\frac{23}{216}$ 3) $\frac{33}{216}$ 4) $\frac{53}{216}$
30. The probability that the graph of $y = 16x^2 + 8(a+5)x - 7a - 5 = 0$, is strictly above the x-axis, If $a \in [-20, 0]$
- 1) $\frac{13}{20}$ 2) $\frac{23}{20}$ 3) $\frac{3}{20}$ 4) $\frac{1}{19}$
31. If p, q are chosen randomly with replacement from the set $\{1, 2, 3, \dots, 10\}$ the probability, that the roots of the equation $x^2 + px + q = 0$ are real is
- 1) $\frac{3}{5}$ 2) $\frac{31}{50}$ 3) $\frac{61}{100}$ 4) $\frac{29}{50}$
32. If 35 fruits are distributed among 3 patients, then the probability, that no one gets less than 10 is
- 1) $\frac{5}{111}$ 2) $\frac{7}{111}$ 3) $\frac{5}{222}$ 4) $\frac{7}{222}$
33. Eight players $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8$ participate in a boxing ring of a knock out and it is known that if p_i and p_j fight p_i will win ($i < j$), assume players are paired at random probability P_4 reaches finals is.
- 1) $\frac{7}{35}$ 2) $\frac{4}{35}$ 3) $\frac{1}{2}$ 4) 1
34. The decimal parts of the logarithms of two numbers taken at random are found to six places. what is the chance that the second can be subtracted from the first without borrowing?
- 1) $\left(\frac{1}{2}\right)^6$ 2) $\left(\frac{11}{20}\right)^6$ 3) $\left(\frac{11}{19}\right)^6$ 4) $\left(\frac{10}{19}\right)^6$
35. A three digit number is written down at random. The probability that it will have two and only two consecutive identical place values is
- 1) $\frac{2}{81}$ 2) $\frac{3}{50}$ 3) $\frac{9}{50}$ 4) $\frac{7}{50}$
36. India plays two matches each with west indies and Australia. In any match, the probabilities. India getting points 0, 1 and 2 are 0.45, 0.05, 0.50 respectively. Assuming that the out are independent, the probability of india getting atleast 7 points is
- 1) 0.8750 2) 0.0875 3) 0.0625 4) 0.0
37. A box contain N coins, m of which are fair and rest are biased. the probability of getting a head when a fair coin is tossed is $\frac{1}{2}$, while it is $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. the probability that the coin drawn is fair if
- 1) $\frac{8m}{8N+m}$ 2) $\frac{m}{8N+m}$ 3) $\frac{9m}{8N+m}$ 4) $\frac{9N}{8N+m}$
38. Four numbers are multiplied together. Then the probability that the product will be divisible by 5 or 10 is_____
- 1) $\frac{369}{625}$ 2) $\frac{399}{625}$ 3) $\frac{125}{625}$ 4) $\frac{133}{625}$
39. A signal which can be green or red with probability $\left(\frac{4}{5}\right)$ and $\left(\frac{1}{5}\right)$ respectively is received by the station A and Transmitted to B. The probability each station receive signal correctly is $\left(\frac{3}{4}\right)$. If the signal received by B is green, then the probability original signal was green is
- 1) $\frac{3}{5}$ 2) $\frac{6}{7}$ 3) $\frac{20}{23}$ 4) $\frac{9}{20}$
40. Let $w \neq 1$ be a cube root of unity. A fair die is rolled three times. If r_1, r_2 and r_3 are the numbers obtained on the die, then the probability that $w^{r_1} + w^{r_2} + w^{r_3} = 0$ is
- (AIEEE 2010)
- 1) $\frac{1}{18}$ 2) $\frac{1}{9}$ 3) $\frac{2}{9}$ 4) $\frac{1}{36}$
41. A car is parked by a driver amongst 25 cars in a row, not at either end. When he returns he finds that 10 places are empty. The probability that both the neighbouring places of drivers car are vacant is
- 1) $\frac{9}{92}$ 2) $\frac{15}{92}$ 3) $\frac{21}{92}$ 4) $\frac{27}{92}$

42. If four dice are thrown together, then the probability that the sum of the numbers appearing on them is 13 is

1) $\frac{5}{216}$ 2) $\frac{11}{216}$ 3) $\frac{35}{324}$ 4) $\frac{11}{432}$

LEVEL - III - KEY

- 1) 1 2) 1 3) 2 4) 3 5) 4 6) 1 7) 4
 8) 3 9) 3 10) 3 11) 1 12) 1 13) 3 14) 3
 15) 2 16) 3 17) 3 18) 1 19) 2 20) 2 21) 3
 22) 3 23) 2 24) 3 25) 3 26) 4 27) 4 28) 2
 29) 1 30) 1 31) 2 32) 4 33) 2 34) 2 35) 3
 36) 2 37) 3 38) 1 39) 3 40) 3 41) 2 42) 3

LEVEL - III - HINTS

- $n(S) = \frac{100!}{50!50!2!}$ $n(A) = \frac{2.98!}{49!49!2!}$ $P(A) = \frac{50}{99}$
- $x = 33^n$; only four digits may be at the units place i.e., 1, 3, 7, 9; $n(S) = 4$
 Let A be the event of getting 3 at its units place
 $n(A) = 1$; $P(A) = 1/4$
- $n(S) = 5^3 = 125$
 A function $f: A \rightarrow B$ such that $f(1) \leq f(2) \leq f(3)$ falls in one of the following four categories
Case 1 : $f(1) < f(2) < f(3)$
 No. of trails in this case $5C_3$
Case 2 : $f(1) = f(2) < f(3)$
 No. of trails in this case $5C_2$
Case 3 : $f(1) < f(2) = f(3)$
 No. of trails in this case $5C_2$
Case 4 : $f(1) = f(2) = f(3)$
 No. of trails in this case $5C_1$
 No. of desired trails $n(E) = 35$; $P(E) = 7/25$
- $\{(B_1, G_1) (B_2, G_2) (B_3, G_3) (B_4, G_4) B_5, B_6\}$
 Req. Probability =
 $1 - \frac{{}^4C_1 \times 8}{{}^{10}C_3} = 1 - \frac{4}{15} = \frac{11}{15}$
- Let A be the event of getting even no and B be the event of getting an odd number on a die.
 $P(A) = \frac{5}{18}$ $P(B) = \frac{1}{18}$
 $P(2 \text{ or } 3 \text{ or } 5 \text{ or } 7 \text{ or } 11) =$
 $\frac{1}{18} \cdot \frac{1}{18} + 2 \cdot \frac{1}{18} \cdot \frac{5}{18} + 4 \cdot \frac{1}{18} \cdot \frac{5}{18} + 6 \cdot \frac{1}{18} \cdot \frac{5}{18} + 2 \cdot \frac{1}{18} \cdot \frac{5}{18}$
 $= \frac{71}{324}$

6. ${}^6C_2 - {}^4C_2 = 9$; $n(S) = {}^9C_2 = 36$,
 no. of selections of four vertices $= 6C_4 = 15$;

$$P(E) = \frac{5}{12}$$

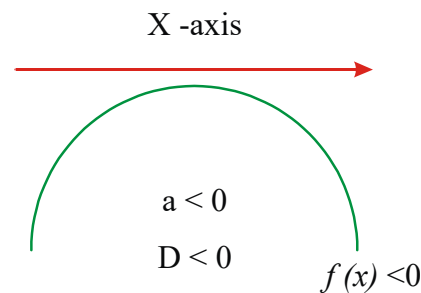
7. Error is at least 15 paise when we round off 15P,
 16P.....49P, 50P, 51P.....85P, $P(E) = \frac{71}{100}$

8. $7^1 = 7$, $7^2 = 49$, $7^3 = 343$, $7^4 = 2401$

From each of value of m in $\{1, 2, 3, 4, \dots, 100\}$ there are 25 values of n

$$\text{Req. prob} = \frac{100 \times 25}{100 \times 100} = \frac{1}{4}$$

9. The total length of the interval $= 12 - (-6) = 18$. If graph of $y = -x^2 + 2(a+4)x - (3a+40)$ is entirely below x-axis, the value of discriminant of the above quadratic expression must be negative



$$\therefore 4(a+4)^2 - 4(-1)(-3a+40) < 0$$

$$\Rightarrow a^2 + 5a - 24 < 0 \Rightarrow (a+8)(a-3) < 0 \Rightarrow -8 < a < 3$$

but $a \in [-6, 12]$ $\therefore -6 < a < 3$ for event to happen
 $\therefore$ length of interval $= 3 - (-6) = 9$

$$\text{Hence of required probability} = \frac{9}{18} = 1/2$$

- $n(s) = 7^4$, $n(A) = 3^4 - 2^4$
- $n(s) = 10^4$, $n(A) = 4^4$
- 'x' should be a multiple of the L.C.M of 2, 3, 5 (i.e., 30) $\therefore n(A) = 33$
- $P(\overline{A}) : P(A) = 2 : 1$; $P(A \cup B) : P(\overline{A \cup B}) = 3 : 1$
 $P(A \cup B) - P(A) \leq P(B) \leq P(A \cup B)$
 $\frac{5}{12} \leq P(B) \leq \frac{3}{4}$
- $P(A) = 1 - \left(\frac{86}{90}\right)^3 = 1 - \left(\frac{43}{45}\right)^3$

$$15. RP = \frac{\frac{4}{5} \times \frac{3}{4}}{\frac{4}{5} \times \frac{3}{4} + \frac{1}{5} \times \frac{1}{4}}$$

$$16. R.P = \frac{1}{n-1}$$

17. Impossible event

$$18. R.P = 1 - \frac{\pi a^2 (1/3)}{\pi a^2}$$

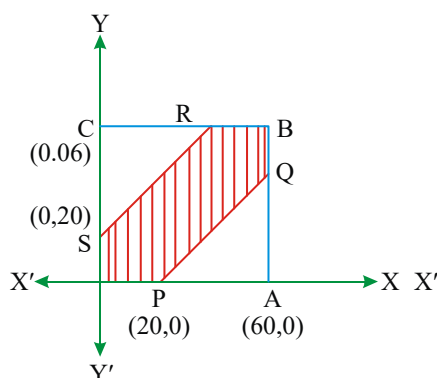
19. Let A and B arrive at the place of their meeting x minutes and y minutes after 11 noon.

The given condition $\Rightarrow$ their meeting is possible only if $|x - y| \leq 20$ 1

OABC is a square, where $A \equiv (60, 0)$ and

$C \equiv (0, 60)$ Considering the equality part of (1)

i.e., $|x - y| = 20$



Probability

$\therefore$ The area representing the favourable cases

= Area OPQBSO

= Area of square OABC - Area of Δ PAQ - Area of Δ SRC

$$= (60)(60) - \frac{1}{2}(40)(40) - \frac{1}{2}(40)(40)$$

$$= 3600 - 1600 = 2,000 \text{ sq. units}$$

Total way = Area of square OABC

$$= (60)(60) = 3,600 \text{ sq. units}$$

$$\therefore \text{ Required probability} = \frac{2,000}{3,600} = \frac{5}{9}$$

(or) Take $a = 11$, $b = 12$, $c = 20$ min $= 1/3$ hr

$$\text{Required probability} = \frac{(b-a)^2 - (1-c)^2}{(b-a)^2}$$

$$= \frac{(12-11)^2 - \left(1 - \frac{1}{3}\right)^2}{(12-11)^2} = 1 - \frac{4}{9} = \frac{5}{9}$$

20. Total number of ways choosing x & y is

$${}^{3n}C_2 = \frac{3n(3n-1)}{2} \text{ If S be the sample space}$$

$$\text{then } n(S) = \frac{3n(3n-1)}{2} \text{ Now arrange the given}$$

number as below

$$1, 4, 7, \dots, 3n-2$$

$$2, 5, 8, \dots, 3n-1$$

$$3, 6, 9, \dots, 3n$$

We see that $x^3 + y^3$ will be divisible by 3 in the following cases

one of two numbers belongs to the first row; one of the two numbers belongs to the second row; or both numbers occurs in third row. If E be the event for favourable cases, then

$$n(E) = (n)(n) + {}^n C_2 = n^2 + \frac{n(n-1)}{2}$$

$$= \frac{n}{2}(3n-1)$$

$$\text{Required probability} = \frac{\frac{n}{2}(3n-1)}{\frac{n}{2}(3n-1)} = \frac{1}{3}$$

$$\text{Probability} = \frac{\text{Area of } E}{\text{Area of } S} = \frac{3s^2}{4s^2} = \frac{3}{4}$$

21. The number of ways for the numbers x, y to have digit each in their units place $= 10 \times 10 = 100$.

The number of ways for the numbers x, y to have a digit in their units places whose sum is 0 or 10 $= 1 \times 1 + 2 \times 2 + 2 \times 2 + 1 \times 1 + 2 \times 2 + 2 \times 2 = 18$

$$\text{the required probability is} = \frac{18}{100} = \frac{9}{50}$$

22. Since sum of 4 digit number is 34.

It must contain atleast two '9's since $4 \times 8 = 32$ So, the 4 digits of the number will be as follows.

(i) 9, 9, 8, 8 (ii) 9, 9, 9, 7

The total number of 4-digit numbers

whose sum is $34 = \frac{4}{2} + \frac{4}{3} = 6 + 4 = 10$

Among these 10 numbers,

4-digit numbers ended with 8 = $\frac{3}{2} = 3$

∴ The number of 4 digit even numbers with sum of digits 34 = 3

The required probability = $\frac{3}{10}$

$$23. n(A) = 2 \left[{}^{100}C_1 + {}^{100}C_3 + \dots + {}^{100}C_{49} \right] = 2^{99}$$

$$\text{Required probability} = \frac{2^{98}}{2^{100}} = \frac{1}{4}$$

$$24. \text{Sol: } n(S) = {}^{25}C_2 = 300$$

$$n(E) = 9 \times 16 + \sum 8 = 144 + 36 = 180$$

$$25. e^{\lim_{x \rightarrow 0} \frac{2 \left(\frac{a^x + b^x}{2} - 1 \right)}{x}} = 6$$

$$\Rightarrow ab = 6$$

$$(a, b) = \{(1, 6), (6, 1), (2, 3), (3, 2)\}$$

$$\text{probability} = \frac{4}{6 \times 6} = \frac{1}{9}$$

$$26. n(S) = 10, n(A) = 4, n(B) = x$$

$$P(A \cap B) = P(A)P(B)$$

$$\frac{y}{10} = \frac{4x}{100}$$

$$27. P(E) = \frac{\frac{1}{2}(1)}{\frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot \frac{1}{8} + \frac{1}{2} \cdot 1} = \frac{24}{29}$$

28. Let C, S, B and T be the events of Ramesh using car, scooter, bus or train.

Let L be the event of Ramesh reaching office late.

By hypothesis

$$P(C) = \frac{1}{7}, P(S) = \frac{3}{7}, P(B) = \frac{2}{7} \text{ and } P(T) = \frac{1}{7}$$

$$P\left(\frac{L}{C}\right) = \frac{2}{9}, P\left(\frac{L}{S}\right) = \frac{1}{9}, P\left(\frac{L}{B}\right) = \frac{4}{9}, P\left(\frac{L}{T}\right) = \frac{1}{9}$$

$$P\left(\frac{\bar{L}}{C}\right) = 1 - \frac{2}{9} = \frac{7}{9}, P\left(\frac{\bar{L}}{S}\right) = 1 - \frac{1}{9} = \frac{8}{9}$$

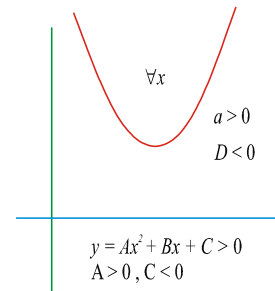
$$P\left(\frac{\bar{L}}{B}\right) = 1 - \frac{4}{9} = \frac{5}{9}, P\left(\frac{\bar{L}}{T}\right) = 1 - \frac{1}{9} = \frac{8}{9}$$

By baye's theorem,

$$P\left(\frac{C}{\bar{L}}\right) = \frac{P(C)P\left(\frac{\bar{L}}{C}\right)}{P(C)P\left(\frac{\bar{L}}{C}\right) + P(S)P\left(\frac{\bar{L}}{S}\right) + P(B)P\left(\frac{\bar{L}}{B}\right) + P(T)P\left(\frac{\bar{L}}{T}\right)}$$

$$29. n(E) = 1 + 3 + 8 + 14 + 17 = 43, n(S) = 216.$$

$$30. y > 0 \forall x \in R$$



$$\Rightarrow 64(a+5)^2 + 4.16(7a+5) < 0$$

$$\Rightarrow a^2 + 17a + 30 < 0 \Rightarrow (a+15)(a+2) < 0$$

$$31. \begin{array}{ll} p & q \\ 2 & 1 \\ 3 & 1, 2 \\ 4 & 1 \text{ to } 4 \\ 5 & 1 \text{ to } 6 \\ 6 & 1 \text{ to } 9 \\ 7, 8, 9, 10 & 1 \text{ to } 10 \end{array}$$

The total number of pairs (p, q) is $1 + 2 + 4 + 6 + 9 + 40 = 62, n(E) = 62, n(S) = 100.$

$$32. x_1 + x_2 + x_3 = 35, n(E) = 21, n(S) = 37 \times 18$$

$$P(E) = \frac{21}{18 \times 37} = \frac{7}{6 \times 37} = \frac{7}{222}$$

$$33. n(E) = 4, n(S) = {}^7C_3$$

34. Each position subtraction can be
 $(0, 0), (1, 0) \dots \dots \dots (9, 0) \rightarrow 10$ ways
 $(1, 1), (2, 1) \dots \dots \dots (9, 1) \rightarrow 9$ ways
 so on $\dots \dots \dots (9, 9) \rightarrow 1$ ways

$$p(E) = \left(\frac{55}{100} \right)^6$$

$$35. n(E) = 81 + 81 = 162; n(S) = 9 \times 10 \times 10 = 900$$

$$36. P(X \geq 7) = P(X = 7) + P(X = 8) \\ = 4 \times (0.50)^3 (0.05) + (0.50)^4$$

37. First H and secondly T, coin drawn fair,

$$P\left(\frac{B}{A}\right) = \frac{P(A \cup B)}{P(A)} = \frac{\frac{m}{n} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{m}{n} \cdot \frac{1}{2} \cdot \frac{1}{2} + \left(\frac{N-m}{N}\right) \frac{2}{3} \cdot \frac{1}{3}} = \frac{9m}{8N+m}$$

38. Required probability = 1 - p(not divisible by 5 or 10)

$$= 1 - \left(\frac{8}{10}\right)^4 = \frac{5904}{1000} = \frac{369}{625}$$

39. G = Original signal green. A = A receive correct signal, B = B receive signal correct. E is signal received by B is green.

$$P\left(\frac{G}{E}\right) = \frac{P(G \cap E)}{P(E)}$$

$$P(E) = P(GAB) + P(G\bar{A}\bar{B}) + P(\bar{G}A\bar{B}) + P(\bar{G}\bar{A}B)$$

$$= \frac{4}{5} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{46}{80}$$

$$P(G \cap E) = P(GAB) + P(G\bar{A}\bar{B}) = \frac{40}{80}$$

40. Let E be the set of ordered triads (r_1, r_2, r_3) such that $w^{r_1} + w^{r_2} + w^{r_3} = 0$ It is known that $w^3 = 1, w^4 = w, w^5 = w^2, w^6 = 1$ and

$$1 + w + w^2 = 0$$

$$E = \{(3, 1, 2), (3, 1, 5), (3, 4, 2), (3, 4, 5),$$

$$(6, 1, 2), (6, 1, 5), (6, 4, 2), (6, 4, 5)\}$$

E contains 8×6 sample points. $\therefore P(E) = \frac{8 \times 6}{6^3} = \frac{2}{9}$

41. Out of 24 places (except driver's car) 10 are vacant, 10 vacant places out of 24 can be selected in

$${}^{24}C_{10} \text{ ways } \therefore n(s) = {}^{24}C_{10}$$

If neighbouring places on either side of drivers car are vacant then out of remaining 22 places 8 are vacant which can be selected in ${}^{22}C_8$ ways.

$$\text{Required probability} = \frac{{}^{22}C_8}{{}^{24}C_{10}} = \frac{15}{92}$$

42. Keeping 1, 2, 3, 4, 5, 6 on first die, count the number of cases using table (formula) for the remaining sum on three dice. $n(A) = 140; n(s) = 6^4$



LEVEL - IV

1. A number 'n' is chosen at random from $S = \{1, 2, 3, \dots, 50\}$. Let

$$A = \left\{n \in S : n + \frac{50}{n} > 27\right\},$$

$$B = \{n \in S : n \text{ is a prime}\} \text{ and}$$

$C = \{n \in S : n \text{ is a square}\}$. The correct order of their probabilities is

- 1) $P(A) < P(B) < P(C)$ 2) $P(A) > P(B) > P(C)$
3) $P(B) < P(A) < P(C)$ 4) $P(A) > P(C) > P(B)$

2. An unbiased coin is tossed to get 2 points for turning up a head and one point for the tail. If three unbiased coins are tossed simultaneously, then the probability of getting

I : a total of odd no. of points is $\frac{2}{3}$

II : a total of 4 or 5 is $\frac{3}{4}$

Which of the above statements is / are correct

- 1) I only 2) II only
3) both I and II 4) neither I nor II

3. A problem is given to two students A and B

whose chances of solving it are $\frac{1}{3}, \frac{1}{5}$ respectively then

LIST-I

LIST-II

A) probability that the problem is solved is

1) $\frac{7}{15}$

B) probability that the problem not solved is

2) $\frac{4}{15}$

C) probability that the problem solved by A only is

3) $\frac{6}{15}$

D) probability that the problem solved by B only is

5) $\frac{8}{15}$

The correct match for LIST-I from LIST-II is

	A	B	C	D
1)	2	1	5	4
2)	3	5	1	4
3)	1	5	2	4
4)	2	1	5	3

4. A, B are two events in a random experiment such that $0 < P(A) < 1$ and $P(B) \neq 1$.

$$\text{Assertion (A) :- } P\left(\frac{A}{B^c}\right) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

$$\text{Reason (R) :- } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

- 1) both A and R are true and R is the correct explanation of A
 2) both A and R are true and R is not the correct explanation of A
 3) A is true but R is false
 4) A is false but R is true

5. If A, B, C are three independent events of an experiment such that $P(A \cap B^c \cap C^c) = \frac{1}{4}$,

$$P(A^c \cap B \cap C^c) = \frac{1}{8}, P(A^c \cap B^c \cap C^c) = \frac{1}{4} \text{ then the}$$

increasing order of $P(A)$, $P(B)$ and $P(C)$

- 1) $P(C) < P(B) < P(A)$ 2) $P(B) < P(A) < P(C)$
 3) $P(A) < P(C) < P(B)$ 4) $P(B) < P(C) < P(A)$

6. The probability that Australia wins a match against India in a cricket match is $\frac{1}{3}$. India and Australia play 3 matches

I. The probability that Australia wins atleast one match is $\frac{4}{27}$

II : The probability that Australia loses all the three matches is $\frac{8}{27}$

Which of the above statements is correct

- 1) I only 2) II only
 3) both I and II 4) neither I nor II

7. I : A single die is rolled twice in succession. The probability that the number showing on the second toss is greater than that on the first rolling is $\frac{4}{27}$.

II : The probability that a boy gets a scholarship is 0.8 and another boy B is 0.9. the probability that atleast one of them gets the scholarship is 0.98. Which of the above statements is true

- 1) I only 2) II only
 3) both I and II 4) neither I nor II

8. A bag X contains 4 white and 2 black balls. Another bag Y contains 3 white and 4 black balls. A bag is selected at random and a ball is drawn at random from it.

I. the probability that the ball drawn is white is $\frac{17}{42}$

II. The probability that the ball drawn is black is $\frac{19}{42}$

Which of the above statements is correct

- 1) only I 2) only II
 3) both I and II 4) neither I nor II

9. An electrical shopkeeper purchases bulbs from three manufacturers A, B, C. He purchases 25% of his stock from A, 45% from B and 30% from C. In the past purchases he found 2% of C's bulbs, 1% of B's bulbs, 1% of A's bulbs were defective. Assuming that, the shopkeeper chooses a bulb and found it to be defective

LIST-I

LIST-II

- A) The probability of choosing A and purchasing a defective bulb from it is 1) 0.0025
 B) The probability of choosing B and purchasing a defective bulb from it is 2) 0.0045
 C) The probability of choosing C and purchasing a defective bulb from it is 3) 0.0060
 D) The probability that the defective bulb was purchased from C is 4) 0.65
 5) 0.46

The correct Match for LIST-I from LIST-II is

	A	B	C	D
1)	4	5	2	3
2)	5	2	3	1
3)	2	3	1	5
4)	1	2	3	5

10. Two events A and B have probabilities 0.25 and 0.50 respectively. The probability that both A and B occur simultaneously is 0.14. p = the probability that exactly one of A and B occurs; q = the probability that neither A or B occurs; r = the probability that A occurs and B does not occur. Which of the following is true

- 1) $2q < 5r < p$ 2) $p < 5r < q$
 3) $4r < p < q$ 4) $p < 5r < 2q$

11. Two dice are thrown p = the probability of getting a number always greater than 4 on the second die; q = the probability of getting a sum 10 on the two dice and r = the probability of getting a sum which is a perfect square. Arrange the probabilities in increasing order of magnitude

1) p, q, r 2) q, r, p 3) r, p, q 4) r, q, p

12. p = probability for a leap year to have 52 Mondays and 52 Wednesdays; q = probability for a non-leap year to have 53 Sundays; r = probability of the month February in a leap year to have 5 Sundays or 5 Saturdays. Then

1) $p < q < r$ 2) $q < r < p$ 3) $r < q < p$ 4) $q < p < r$

13. A box contains 2 red, 3 blue and 4 black balls, 3 balls are drawn at random from the box. p = the probability for the balls to be of different colours; q = the probability for 2 balls of the same colour and the third of different colour; r = the probability for all the 3 balls to be of the same colour. Arrange them in the decreasing order of magnitude

1) p, q, r 2) q, r, p 3) q, p, r 4) r, q, p

14. A and B be two independent events of a sample space such that $P(A) = 0.2$, $P(B) = 0.5$

LIST-I

A) $P(B/A)$

B) $P(A/B)$

C) $P(A \cap B)$

D) $P(A \cup B)$

LIST-II

1) 0.2

2) 0.1

3) 0.3

4) 0.6

5) 0.5

The correct match for List-I from List-II

	A	B	C	D
1)	4	5	3	1
2)	5	1	2	4
3)	3	1	2	4
4)	2	1	4	5

15. A, B are two events of a sample space such

that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{2}{15}$

LIST-I

A) $P(A/B) =$

B) $P(B/A) =$

C) $P(A^1/B^1) =$

D) $P(B^1/A^1) =$

LIST-II

1) $3/5$

2) $1/2$

3) $4/15$

4) $2/5$

5) $9/20$

The correct match for LIST-I from LIST-II is

	A	B	C	D
1)	4	3	2	1
2)	5	4	3	2
3)	4	3	2	5
4)	3	4	1	5

16. A husband and wife appear in interview for the same post. The probability of husband's selection is $1/7$ and that of wife's selection is $1/5$. Let a = prob. that both of them will be selected; b = Prob. that none of them will be selected and c = Prob. that only one of them will be selected. Then we have

1) $a < c < b$ 2) $a < b < c$ 3) $b < a < c$ 4) $c < a < b$

17. A lot contains 50 defective and 50 non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as $A = \{\text{the first bulb is defective}\}$, $B = \{\text{the second bulb is non defective}\}$, $C = \{\text{the two bulbs are both defective or both non defective}\}$. Then

(i) A, B, C are pair wise independent

(ii) A, B, C are independent

1) only (i) is true

2) (i) and (ii) are true

3) only (ii) is true

4) (i) and (ii) are false

18. Three boxes A, B, C contain balls as shown below

	White	Red	Black
A	2	2	1
B	3	4	2
C	4	2	3

A die is thrown to decide which box is to be chosen. A is chosen if 1 or 2 turns up; B is chosen if 3 or 4 turns up; C is chosen if 5 or 6 turns up. Having chosen the box, a ball is chosen at random from it

I. Probability for the ball to be red is $16/44$

II : If the drawn ball is of red colour, the probability that it is from box B is $5/8$.

Which of the above statement(s) is/are correct

1) I only

2) II only

3) both I and II

4) neither I nor II

19. An urn A contains 2 white and 3 black balls. Another urn B contains 3 white and 4 black balls. Out of these two urns, one is selected at random and a ball is drawn from it. If the ball drawn is black, then the probability

I. that it is from urn A is $21/40$

II. that is from urn B is $20/41$

Which of the following statements is correct

1) I only

2) II only

3) both I and II

4) neither I nor II

20. By rolling a die the following events are considered. $E_1 = \{5, 3, 1\}$, $E_2 = \{3, 2\}$, $E_3 = \{5, 4, 3, 2\}$. The arrangement of the following probabilities

$$A : P(E_1 / E_2) \quad B : P(E_1 \cup E_3)$$

$$C : P(E_1 \cap E_3) \quad D : P(E_2 \cap E_1)$$

in the descending order is

- 1) B, A, C, D 2) D, C, A, B
3) C, B, A, D 4) B, A, D, C

21. Consider the system of equations

$$ax + by = 0, cx + dy = 0 \text{ where } a, b, c, d \in \{0, 1\}$$

Statement - I The probability that the system of equations has a unique solution is $3/8$.

Statement - II The probability that the system has a solution is 1.

- 1) Statement - I and Statement - II are true
Statement - II is not a correct explanation for statement - I
2) Statement - I and Statement - II are true
Statement - II is correct explanation for statement - I
3) Statement - I is true Statement - II is not true
4) Statement - I is not true Statement - II is true.

LEVEL - IV - KEY

- 1) 2 2) 2 3) 3 4) 1 5) 1 6) 2 7) 2
8) 2 9) 4 10) 4 11) 2 12) 2 13) 3 14) 2
15) 4 16) 1 17) 1 18) 4 19) 2 20) 1 21) 1

LEVEL - IV - HINTS

1. $P(A) = \frac{26}{50}$; $P(B) = \frac{15}{50}$; $P(C) = \frac{7}{50}$
 $P(A) > P(B) > P(C)$
2. $p(\text{two heads, one tail}) + p(\text{two tails one head}) = \frac{3}{4}$
3. $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{5}$
A) $P(A \cup B) = \frac{1}{3} + \frac{1}{5} - \frac{1}{15} = \frac{8-1}{15} = \frac{7}{15}$
B) $1 - P(A \cup B) = \frac{8}{15}$
C) $P(A \cap \bar{B}) = \frac{1}{3} \times \frac{4}{5} = \frac{4}{15}$ D) $P(\bar{A} \cap B) = \frac{2}{3} \times \frac{1}{5} = \frac{2}{15}$

$$4. \quad 0 < P(A) < 1$$

$$P(B) \neq 1$$

$$A : P\left(\frac{A}{B^c}\right) = \frac{P(A \cap \bar{B})}{1 - P(B)} = \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

$$R : P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$5. \quad P(A) \cdot P(B^c) \cdot P(C^c) = \frac{1}{4}$$

$$P(A^c) \cdot P(B) \cdot P(C^c) = \frac{1}{8}$$

$$P(A^c) \cdot P(B^c) \cdot P(C^c) = \frac{1}{4}$$

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(C) = \frac{1}{4}$$

$$P(C) < P(B) < P(A)$$

$$6. \quad I : 1 - \left(\frac{2}{3}\right)^3 = 1 - \frac{8}{27} = \frac{19}{27} \quad II : P(E) = \frac{8}{27}$$

$$7. \quad I) P(E) = \frac{15}{36} = \frac{5}{12} \quad II) P(A) = 0.8$$

$$P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(A \cap B)$$

$$P(B) = 0.9$$

$$= \frac{8}{10} \times \frac{1}{10} + \frac{2}{10} \times \frac{9}{10} + \frac{8}{10} \times \frac{9}{10} = \frac{98}{100} = 0.98$$

$$8. \quad I : R.P \text{ is } = \frac{1}{2} \left[\frac{{}^4C_1}{{}^6C_1} + \frac{{}^3C_1}{{}^7C_2} \right]$$

$$II : R.P \text{ is } \frac{1}{2} \left[\frac{{}^2C_1}{{}^6C_1} + \frac{{}^4C_1}{{}^7C_1} \right]$$

$$9. \quad P(A) = \frac{25}{100}, P(B) = \frac{45}{100}, P(C) = \frac{30}{100}$$

$$P\left(\frac{E}{C}\right) = \frac{2}{100}, P\left(\frac{E}{B}\right) = \frac{1}{100}, P\left(\frac{E}{A}\right) = \frac{1}{100}$$

$$A) P(A \cap E) = P(A) \cdot P\left(\frac{E}{A}\right) = \frac{25}{10000} = 0.0025$$

$$B) P(B \cap E) = 0.0045$$

$$C) P(C \cap E) = 0.0060$$

$$D) \text{ Required probability} = 0.46$$

10. $P(A) = 0.25, P(B) = 0.50$

$$P(A \cap B) = 0.14$$

$$P = P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= (0.25)(0.50) + (0.75)(0.50)$$

$$= \frac{3}{4} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2} = \frac{1}{2}(1)$$

$$q = P(\bar{A} \cap \bar{B}) = \frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$$

$$r = P(A \cap \bar{B}) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

$$\frac{1}{2} < \frac{5}{8} < \frac{6}{8}$$

$$p < 5r < 2q$$

11. $p = \frac{12}{36}, q = \frac{3}{36}, r = \frac{7}{36}$

12. $q < r < p$

$$p = \frac{3}{7}, q = \frac{1}{7}, r = \frac{2}{7}$$

13. $P = \frac{{}^2c_1 \cdot {}^3c_1 \times {}^4c_1}{{}^9c_3}$

$$Q = \frac{{}^2c_2 \times {}^7c_1 + {}^3c_2 \times {}^6c_1 + {}^4c_2 \times {}^5c_1}{{}^9c_3}$$

$$R = \frac{{}^3c_3 + {}^4c_3}{{}^9c_3}$$

14. A) $P\left(\frac{B}{A}\right) = P(B)$

B) $P\left(\frac{A}{B}\right) = P(A)$

C) $P(A \cap B) = P(A) \cdot P(B)$

D) $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

15. By using conditional probability concept.

16. $P(H) = \frac{1}{7}, P(W) = \frac{1}{5}$

$$a = P(H \cap W) = \frac{1}{7} \times \frac{1}{5} = \frac{1}{35}$$

$$b = P(\bar{H} \cap \bar{W}) = 1 - P(H \cup W)$$

$$= 1 - \left[\frac{1}{7} + \frac{1}{5} - \frac{1}{35} \right] = 1 - \left[\frac{7+5-1}{35} \right] = \frac{24}{35}$$

$$c = P(\bar{H} \cap W) + P(H \cap \bar{W})$$

$$= \frac{6}{7} \times \frac{1}{5} + \frac{1}{7} \times \frac{4}{5} = \frac{10}{55}$$

$$a < c < b$$

17. A, B, C are pair-wise independent. A, B, C are not mutually independent

18. I: $P(R) = \frac{1}{3}\left(\frac{2}{5}\right) + \frac{1}{3}\left(\frac{4}{9}\right) + \frac{1}{3}\left(\frac{2}{9}\right)$

$$\text{II: } P\left(\frac{B}{R}\right) = \frac{P(B) \cdot P\left(\frac{R}{B}\right)}{P(R)}$$

19. I: $\frac{{}^3c_1}{{}^5c_1} \cdot \frac{{}^4c_1}{{}^7c_1}$

$$\text{II: } \frac{{}^4c_1}{{}^7c_1} \cdot \frac{{}^3c_1}{{}^5c_1} + \frac{{}^4c_1}{{}^7c_1}$$

20. $A = P\left(\frac{E_1}{E_2}\right) = \frac{1}{2} = \frac{3}{6}$

$$B = P(E_1 \cup E_3) = \frac{3+4-2}{6} = \frac{5}{6}$$

$$C = P(E_1 \cap E_3) = \frac{2}{6}$$

$$D = P(E_2 \cap E_1) = \frac{1}{6}$$

Decreasing order B, A, C, D

21. System of equations have unique solutions

$$\text{if } \Delta \neq 0 \Rightarrow ac - bc \neq 0$$

$$\Leftrightarrow ad = 1 \text{ and } bc = 0 \text{ or}$$

$$ad = 0 \text{ and } bc = 1$$

Statement - I Probability is $\frac{6}{2^4} = \frac{3}{8}$

Statement - II Probability = $\frac{16}{2^4} = 1$